

Spline Interpolation Method

-Mathematical overview-

Project: Spline Interpolation DSP ASIC
(TSMC 65nm)

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1. Introduction – what is interpolation?

Interpolation is a mathematical technique used to estimate intermediate values between known discrete data points. In digital signal processing, interpolation is a key component in sample-rate conversion, up-sampling, and waveform reconstruction. A desirable interpolator should preserve signal fidelity while maintaining smooth time-domain behavior and good spectral properties.

2. Classical Cubic Spline Interpolation

Classical cubic spline interpolation constructs a global piecewise-cubic function $S(t)$ that passes through all sample points and enforces continuity of the function and its first and second derivatives (C^2 continuity) at all interior knots. Although this approach provides excellent smoothness, it requires solving a global linear system, which is computationally expensive and poorly suited for streaming or hardware implementations.

To estimate a value between two samples $(t_i, x_i), (t_{i+1}, x_{i+1})$ we define a cubic polynomial:

$$(1) S_i(t) = a_i(t - t_i)^3 + b_i(t - t_i)^2 + c_i(t - t_i) + d_i \quad t \in [t_i, t_{i+1}]$$

The global $s(t)$ consists of $s_i(t)$ for each interval with conditions to maintain smoothness

3. Motivation for a Local Method

In real-time and hardware-oriented systems, global spline solvers are impractical due to their high latency, memory requirements, and lack of causality. Therefore, local interpolation methods are preferred. The goal is to approximate the smoothness of a cubic spline while relying only on a small local neighborhood of samples.

4. Cubic Hermite Interpolation

Interpolation is performed using cubic Hermite polynomials. Given two consecutive samples x_i and x_{i+1} , the interpolant on the interval $[t_i, t_{i+1}]$ is defined using the sample values and their associated slopes.

The cubic Hermite form:

$$(2) S_i(t) = H_{00}(u) + H_{01}(u)x_{i+1} + H_{10}(u)hs_i + H_{11}(u)hs_{i+1} \quad t \in [t_i, t_{i+1}]$$

Where:

- $h = t_{i+1} - t_i \quad \forall i$ (constant sampling rate)

- $u = \frac{t-t_i}{h}$ $u \in [0,1]$ (fractional delay)
- s_i, s_{i+1} slopes at t_i, t_{i+1} respectively
- $H_{00}, H_{01}, H_{10}, H_{11}$ are the cubic hermite basis polynomials:

$$H_{00}(u) = 2u^3 - 3u^2 + 1$$

$$H_{01}(u) = -2u^3 + 3u^2$$

$$H_{10}(u) = u^3 - 2u^2 + u$$

$$H_{11}(u) = u^3 - u^2$$

Define the per-sample slope:

$$(3)m_i = h s_i = h \frac{dx}{dt} \big|_{t=t_i}$$

The interpolant at equation (2) becomes:

$$(4)S_i(u) = H_{00}(u)x_i + H_{01}(u)x_{i+1} + H_{10}(u)m_i + H_{11}(u)m_{i+1} \quad u \in [0,1]$$

We use this version to evaluate new interpolated samples independent of time grid.

5. Smoothness Properties

Each Hermite segment is a cubic polynomial and therefore has continuous first and second derivatives within the interval. However, continuity of the second derivative across knot points is not automatically guaranteed when slopes are chosen locally. The interpolant constructed in this manner is globally C^1 but not strictly C^2 .

6. Minimum-Acceleration-Jerk (MinAJ2) Principle

To improve smoothness across knot points, the MinAJ2 method selects slopes by minimizing the accumulated squared jerk (the third derivative of $S(t)$) over adjacent segments. Although jerk minimization does not explicitly enforce second-derivative continuity, it strongly suppresses curvature jumps, resulting in smooth transitions and favorable spectral behavior.

$$\min J = \int (S'''(t))^2 dt$$

7. Local Two-Segment Optimization Problem

Consider three consecutive uniformly spaced samples x_{i-1}, x_i, x_{i+1} at times t_{i-1}, t_i, t_{i+1}

Two cubic segments are constructed: one on $[t_{i-1}, t_i]$ and one on $[t_i, t_{i+1}]$. The only unknown is the slope m_i at the middle knot; x_{i-1}, m_{i-1}, x_i , and x_{i+1} are known.

The objective is to determine the slope m_i at the middle knot that minimizes the total squared jerk over the two segments.

$$(5) \min J(m_i) = \int_{t_{i-1}}^{t_i} (S_{i-1}'''(t))^2 dt + \int_{t_i}^{t_{i+1}} (S_i'''(t))^2 dt$$

Where:

$$S_{i-1}''(t) = \frac{1}{h^2} [(12u - 6)x_{i-1} + (-12u + 6)x_i + (6u - 4)m_{i-1} + (6u - 2)m_i]$$

$$(6) S_{i-1}'''(t) = \frac{6}{h^3} [2(x_{i-1} - x_i) + (m_i + m_{i-1})]$$

$$S_i''(t) = \frac{1}{h^2} [(12u - 6)x_i + (-12u + 6)x_{i+1} + (6u - 4)m_i + (6u - 2)m_{i+1}]$$

$$(7) S_i'''(t) = \frac{6}{h^3} [2(x_i - x_{i+1}) + (m_i + m_{i+1})]$$

Constraints:

$$S_i'(t_i) = S_{i-1}'(t_i)$$

$$S_i''(t_i) = S_{i-1}''(t_i)$$

- Left segment $[t_{i-1}, t_i]$, with $u = \frac{t-t_{i-1}}{h} \in [0,1]$:

$$S_{i-1}''(t) = \frac{1}{h^2} [(12u - 6)x_{i-1} + (-12u + 6)x_i + (6u - 4)m_{i-1} + (6u - 2)m_i]$$

$$(8) S_{i-1}'''(t) = \frac{6}{h^3} [2(x_{i-1} - x_i) + (m_i + m_{i-1})]$$

- Right segment $[t_i, t_{i+1}]$ with $u = \frac{t-t_i}{h} \in [0,1]$:

$$S_i''(t) = \frac{1}{h^2} [(12u - 6)x_i + (-12u + 6)x_{i+1} + (6u - 4)m_i + (6u - 2)m_{i+1}]$$

$$(9)S_i'''(t) = \frac{6}{h^3} [2(x_i - x_{i+1}) + (m_i + m_{i+1})]$$

We have two unknowns (m_i, m_{i+1}) in equations (8) and (9) but we are only solving for one (m_i).

For the local two-segment optimization, C^2 continuity at the middle knot is imposed as an auxiliary constraint to relate m_{i+1} to m_i . The resulting recursive streaming interpolant is generally globally C^1 , rather than a global classical C^2 spline.

- Left segment $[t_{i-1}, t_i]$, with $u = 1$

$$S_{i-1}''(t_i) = \frac{1}{h^2} [6x_{i-1} - 6x_i + 2m_{i-1} + 4m_i]$$

- Right segment $[t_i, t_{i+1}]$, with $u = 0$:

$$S_i''(t_i) = \frac{1}{h^2} [-6x_i + 6x_{i+1} - 4m_i - 2m_{i+1}]$$

Therefore, the local C^2 constraint gives:

$$\begin{aligned} S_i''(t_i) &= S_{i-1}''(t_i) \\ \rightarrow m_{i+1} &= -4m_i - m_{i-1} - 3x_{i-1} + 3x_{i+1} \end{aligned}$$

This yields:

$$(10)S_i'''(t) = \frac{6}{h^3} [(-3x_{i-1} + 2x_i + x_{i+1}) + (-3m_i - m_{i-1})]$$

Jerk Energy

The third derivative of a cubic polynomial is constant: $S_i'''(t)$ (function of known values $m_i, m_{i-1}, x_{i-1}, x_i, x_{i+1}$). The local accumulated squared jerk over the two segments using equations (5), (8) and (10) is:

$$J(m_i) = \int_{t_{i-1}}^{t_i} (S_{i-1}'''(t))^2 dt + \int_{t_i}^{t_{i+1}} (S_i'''(t))^2 dt =$$

$$\frac{36}{h^5} [(2(x_i - x_{i+1}) + (m_i + m_{i+1}))^2 + ((-3x_{i-1} + 2x_i + x_{i+1}) + (-3m_i - m_{i-1}))^2]$$

Substituting the expressions for $S_{i-1}'''(t)$ and $S_i'''(t)$ gives a quadratic function of m_i .

Minimizing $J(m_i)$ with respect to m_i yields a linear equation whose solution is:

$$m_i = \frac{Ax_{i-1} + Bm_{i-1} + Cx_i + Dx_{i+1}}{E}$$

$$A = -11, B = -4, C = 8, D = 3, E = 10$$

The final MinAJ2 slope recursion formula:

$$(11)m_i = \frac{-11x_{i-1} - 4m_{i-1} + 8x_i + 3x_{i+1}}{10}$$

8. Interpretation and Practical Implications

The MinAJ2 method produces an interpolant that is C^2 within each segment and C^1 globally, with minimized curvature variation at knot points. Because it relies only on local samples and fixed coefficients, it is well suited for streaming, fixed-point, and RTL implementations, while achieving interpolation quality close to that of classical cubic splines.